22 Lecture 22 (March 7, 2025, est.): Ostrik's theorem

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Let k be algebraically closed of characteristic $p > 0$. We finish the proof that every symmetric fusion category over k admits a symmetric tensor functor to Ver_p. The argument follows the Frobenius functor and its fusion image, using the lemmas of the preceding lecture.

22.1 Separating the two factors

Assume first that $p > 3$, that $\mathcal{C}$ is Frobenius injective, and that its Frobenius type is Ver_p^+ or Ver_p. Define $\mathcal{C}_s$ to be the full fusion subcategory whose simple objects satisfy

$$\mathrm{Fr}(X) = Y \boxtimes \mathbf{1} \quad \text{or} \quad \mathrm{Fr}(X) = Y \boxtimes \varepsilon, \qquad \varepsilon = L_{p-1}.$$

This is indeed a fusion subcategory: it is the inverse image of $\mathcal{C}^{(1)} \boxtimes \mathrm{SVec}$ under Fr. By naturality, its own Frobenius functor is the restriction of the one on $\mathcal{C}$. Thus it is Frobenius injective and has type Vect or SVec. Lemmas 21.5 and 21.6 imply

$$\dim(\mathcal{C}_s) \neq 0. \tag{22.1}$$

22.1 Lemma. *Under these assumptions,*

$$\mathrm{Fr}(\mathcal{C}) = \langle \mathrm{Fr}(\mathcal{C}_s), \mathbf{1} \boxtimes \mathsf{Ver}_p^+ \rangle \simeq \mathcal{C}_s \boxtimes \mathsf{Ver}_p^+$$

as symmetric fusion categories.

Proof. Both subcategories on the right belong to the Frobenius image, the second by Lemma 21.7. For the reverse inclusion take a simple $X \in \mathcal{C}$. Its image is simple, so write $\mathrm{Fr}(X) = T \boxtimes L_r$. Put

$$\delta_r = \begin{cases} \mathbf{1}, & r \text{ odd}, \\ \varepsilon, & r \text{ even}. \end{cases}$$

Then $L_r \otimes \delta_r$ belongs to Ver_p^+. Full faithfulness and semisimplicity imply that every object in the fusion image is the image of an object: a direct summand is specified by an idempotent, which lifts through a fully faithful functor and splits in the source. Therefore there is a simple $Y \in \mathcal{C}$ with

$$\mathrm{Fr}(Y) = \mathbf{1} \boxtimes (L_r \otimes \delta_r).$$

Now $\mathrm{Fr}(X \otimes Y)$ has the simple summand $T \boxtimes \delta_r$, because L_r is self-dual and $\mathbf{1}$ is a summand of $L_r \otimes L_r$. Lift this summand to a simple Z of $X \otimes Y$. It satisfies $\mathrm{Fr}(Z) = T \boxtimes \delta_r$, hence $Z \in \mathcal{C}_s$, and

$$\mathrm{Fr}(X) \simeq \mathrm{Fr}(Z) \otimes \big(\mathbf{1} \boxtimes (L_r \otimes \delta_r)\big).$$

This proves generation.

It remains to justify that generation is a Deligne product. The two subcategories have only the unit as a common simple: a simple in $\mathrm{Fr}(\mathcal{C}_s)$ has second factor $\mathbf{1}$ or ε, whereas a simple in $\mathbf{1} \boxtimes \mathsf{Ver}_p^+$ has first factor $\mathbf{1}$ and odd-indexed second factor. The only common possibility is the unit. In a symmetric fusion category, two full fusion subcategories $\mathcal{A}, \mathcal{B}$ with this property have fully faithful product functor $\mathcal{A} \boxtimes \mathcal{B} \to \langle \mathcal{A}, \mathcal{B} \rangle$. Indeed, for simple $A_i \in \mathcal{A}$ and $B_i \in \mathcal{B}$, rigidity identifies

$$\mathrm{Hom}(A_1 \otimes B_1, A_2 \otimes B_2) \simeq \mathrm{Hom}(A_2^* \otimes A_1, B_2 \otimes B_1^*).$$

Only the unit can contribute to this Hom-space; it contributes once exactly when $A_1 \simeq A_2$ and $B_1 \simeq B_2$. The product functor is therefore fully faithful and, by generation, essentially surjective. Apply this with $\mathcal{A} = \mathrm{Fr}(\mathcal{C}_s)$ and $\mathcal{B} = \mathbf{1} \boxtimes \mathsf{Ver}_p^+$, and use $\mathrm{Fr}(\mathcal{C}_s) \simeq \mathcal{C}_s$. ∎

22.2 The theorem

22.2 Theorem (Ostrik)**.** *Every symmetric fusion category $\mathcal{C}$ over k admits a k-linear exact symmetric monoidal functor*

$$\mathcal{C} \longrightarrow \mathsf{Ver}_p.$$

Proof. Suppose there is a counterexample and choose one of minimal Frobenius–Perron dimension. Such a minimum exists. To check this point, fix a bound M. A fusion category of dimension at most M has at most M simple objects, because every simple has Frobenius–Perron dimension at least one. Its simple dimensions are at most $\sqrt{M}$, and its fusion multiplicities satisfy $N_{ij}^r \leq d_i d_j / d_r \leq M$. There are therefore only finitely many possible based fusion rings below the bound, and hence finitely many possible Frobenius–Perron dimensions.

By Lemma 21.2, the fusion image of $\mathrm{Fr} : \mathcal{C} \to \mathcal{C}^{(1)} \boxtimes \mathsf{Ver}_p$ has dimension at most that of $\mathcal{C}$. If it had smaller dimension, minimality would give a symmetric tensor functor from that image to Ver_p, and composition would contradict the choice of $\mathcal{C}$. Consequently equality holds and Fr is fully faithful.

If its type is Vect, or SVec when $p > 2$, the lemmas of Lecture 21 make $\mathcal{C}$ nondegenerate. Proposition 21.1 then supplies a functor to Vect or SVec, followed by the standard inclusion in Ver_p. This is a contradiction. These cases include every possibility for $p = 2, 3$.

Thus $p > 3$, and the type must be Ver_p^+ or Ver_p. By (22.1), the category $\mathcal{C}_s$ has a super fiber functor. Compose

$$\mathcal{C} \xrightarrow{\mathrm{Fr}} \mathrm{Fr}(\mathcal{C}) \simeq \mathcal{C}_s \boxtimes \mathsf{Ver}_p^+ \longrightarrow \mathrm{SVec} \boxtimes \mathsf{Ver}_p^+ \simeq \mathsf{Ver}_p.$$

This is the required functor, contradicting the counterexample. Exactness is automatic for an additive functor with semisimple source. ∎

22.3 Remark. This proves the existence statement in [Ost15, Theorem 1.5]. It does not say that $\mathcal{C}$ itself embeds in Ver_p: the functor need not be full. Nor does the theorem assert that every positive-characteristic symmetric tensor category is semisimple. The generalization beyond fusion categories requires Frobenius exactness, which is the subject of the next lecture.

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